RakeFlow - Experiment results

Key experiments (harness, 30 paired replications, 3 d warm-up + 30 d, seeds 1-30, invariants checked)

Single levers vs baseline

case	rakes/day	t/day	cycle h	P90 cycle h	Ghugus terminal h	crew wait h	held at CD h	demurrage Rs/rake	single-line util	crew expiries/100	CSTPS stock min days	delta rakes/day (paired 95% CI)	delta cycle h
Baseline (today)	4.77 +- 0.06	19,028	40.27 +- 0.45	47.13	12.69	4.44	7.54	33,328	21%	0.02	5.03	+0.00 +- 0.00	+0.0 +- 0.0
Exp 1 Crew ordered during loading	5.77 +- 0.03	23,027	33.27 +- 0.21	38.25	12.55	1.77	3.84	33,440	25%	0.00	7.93	+1.00 +- 0.06	-7.0 +- 0.5
Exp 2 One more loop at Ghugus (3 instead of 2)	5.77 +- 0.06	23,036	33.27 +- 0.33	39.13	12.10	4.43	1.67	33,222	25%	0.00	7.82	+1.00 +- 0.09	-7.0 +- 0.6
Exp 3 Rapid loading silo (RLS) at Ghugus	5.12 +- 0.03	20,411	37.48 +- 0.24	43.25	6.43	4.43	10.61	0	22%	0.00	5.99	+0.35 +- 0.07	-2.8 +- 0.5
Exp 4 Line doubling Ghugus-Tadali (Rs 228 cr)	4.88 +- 0.05	19,453	39.37 +- 0.39	45.89	12.37	4.44	7.02	33,259	11%	0.01	5.38	+0.11 +- 0.06	-0.9 +- 0.5
Extra: yard handling at Ghugus halved	5.50 +- 0.06	21,954	34.95 +- 0.37	41.24	10.64	4.43	4.64	33,400	24%	0.00	6.95	+0.73 +- 0.08	-5.3 +- 0.6
Extra: fleet 8 -> 12 rakes	4.80 +- 0.05	19,148	59.89 +- 0.57	69.44	12.68	4.43	26.30	33,269	21%	0.02	5.04	+0.03 +- 0.08	+19.6 +- 0.7

2x2x2 factorial: crew timing x Ghugus loops x loading method

case	rakes/day	t/day	cycle h	P90 cycle h	Ghugus terminal h	crew wait h	held at CD h	demurrage Rs/rake	single-line util	crew expiries/100	CSTPS stock min days	delta rakes/day (paired 95% CI)	delta cycle h
crew=ready loops=2 loading=CHP	4.77 +- 0.06	19,028	40.27 +- 0.45	47.13	12.69	4.44	7.54	33,328	21%	0.02	5.03	+0.00 +- 0.00	+0.0 +- 0.0
crew=ready loops=2 loading=RLS	5.12 +- 0.03	20,411	37.48 +- 0.24	43.25	6.43	4.43	10.61	0	22%	0.00	5.99	+0.35 +- 0.07	-2.8 +- 0.5
crew=ready loops=3 loading=CHP	5.77 +- 0.06	23,036	33.27 +- 0.33	39.13	12.10	4.43	1.67	33,222	25%	0.00	7.82	+1.00 +- 0.09	-7.0 +- 0.6
crew=ready loops=3 loading=RLS	6.74 +- 0.02	26,897	28.43 +- 0.09	32.45	6.45	4.43	2.44	1	29%	0.00	9.72	+1.97 +- 0.06	-11.8 +- 0.4
crew=loading loops=2 loading=CHP	5.77 +- 0.03	23,027	33.27 +- 0.21	38.25	12.55	1.77	3.84	33,440	25%	0.00	7.93	+1.00 +- 0.06	-7.0 +- 0.5
crew=loading loops=2 loading=RLS	6.76 +- 0.03	26,981	28.38 +- 0.16	33.18	6.42	1.76	5.01	3	29%	0.00	9.85	+1.99 +- 0.07	-11.9 +- 0.5
crew=loading loops=3 loading=CHP	6.37 +- 0.04	25,425	30.10 +- 0.16	35.07	12.24	1.77	1.15	33,299	28%	0.00	9.29	+1.60 +- 0.06	-10.2 +- 0.5
crew=loading loops=3 loading=RLS	7.76 +- 0.04	30,971	24.73 +- 0.11	28.56	6.40	1.76	1.67	0	34%	0.00	10.36	+2.99 +- 0.06	-15.5 +- 0.4

Factorial effects on rakes/day (effect = mean(high) - mean(low))

effect	rakes/day
Crew during loading	+1.07
Loop +1	+1.06
RLS	+0.92
Crew x Loop	-0.26
Crew x RLS	+0.27
Loop x RLS	+0.25
Crew x Loop x RLS	-0.06

RakeFlow harness results

30 replications per case, each 3 days warm-up + 30 days statistics, seeds 1-30 shared by every case (common random numbers). Mean +- 95% CI half-width (t, 29 df). Delta = paired difference in rakes/day against the phase-3 baseline (same seeds).

Phase build-up

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
P1 core cycle (no crew, no BPC)	5.75 +- 0.03	33.4 +- 0.2	37.4	25%	0.05	0.033	100%	26%	0.14	33,200	0.00	0.0	0.0%	7.7	+0.98 +- 0.06
P2 + crew duty clock	4.81 +- 0.04	39.9 +- 0.4	46.0	21%	0.09	0.026	81%	22%	0.13	33,273	0.00	0.0	0.0%	5.0	+0.04 +- 0.07
P3 + BPC exam detours (baseline)	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00

E1 fleet size (A-18)

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
fleet 4	3.07 +- 0.02	31.2 +- 0.2	35.5	13%	0.03	0.017	42%	14%	0.13	33,682	0.00	22.4	4.1%	0.7	-1.70 +- 0.06
fleet 6	4.30 +- 0.03	33.4 +- 0.2	38.8	19%	0.08	0.024	63%	19%	0.13	33,367	0.01	23.0	4.6%	3.9	-0.46 +- 0.06
fleet 8	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
fleet 10	4.83 +- 0.05	49.6 +- 0.5	57.7	21%	0.08	0.025	81%	22%	0.12	33,374	0.00	24.3	6.8%	5.5	+0.06 +- 0.08
fleet 12	4.80 +- 0.05	59.9 +- 0.6	69.4	21%	0.09	0.026	80%	21%	0.13	33,269	0.02	25.7	8.5%	5.0	+0.03 +- 0.08
fleet 14	4.79 +- 0.04	70.0 +- 0.6	81.3	21%	0.09	0.027	81%	22%	0.14	33,396	0.03	26.0	10.0%	5.3	+0.02 +- 0.08
fleet 16	4.86 +- 0.04	79.0 +- 0.7	91.8	21%	0.09	0.026	79%	22%	0.13	33,242	0.00	26.5	11.3%	5.3	+0.09 +- 0.06

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
fleet 18	4.87 +- 0.04	88.6 +- 0.9	103.3	21%	0.10	0.026	79%	22%	0.13	33,064	0.02	27.3	12.6%	5.3	+0.10 +- 0.06
fleet 20	4.84 +- 0.05	98.9 +- 1.1	115.0	21%	0.09	0.028	79%	22%	0.14	33,190	0.01	26.8	13.8%	5.4	+0.07 +- 0.08

E2-E7 levers (one at a time vs baseline)

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
baseline	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
E2 RLS/silo loading (LD-04)	5.12 +- 0.03	37.5 +- 0.2	43.3	22%	0.20	0.029	24%	23%	0.14	0	0.00	23.3	5.1%	6.0	+0.35 +- 0.07
E3 planned DN slot + crew to match	4.78 +- 0.05	40.2 +- 0.4	47.0	21%	0.10	0.026	79%	21%	0.13	33,245	0.03	24.0	5.5%	5.0	+0.01 +- 0.08
E3 priority to crews < 2 h left	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
E4 crew booked at Chandrapur	5.14 +- 0.03	37.4 +- 0.2	43.4	22%	0.05	0.029	90%	23%	0.14	33,334	0.00	23.9	5.1%	6.2	+0.37 +- 0.07
E5 Tadali-Ghugus doubling	4.88 +- 0.05	39.4 +- 0.4	45.9	11%	0.07	0.026	79%	22%	0.13	33,259	0.01	23.3	5.4%	5.4	+0.11 +- 0.06
E6 tipplers 3	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	29%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
E6 tipplers 5	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	17%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
E7 loops +1 at GGS and TAE	5.71 +- 0.07	33.7 +- 0.5	41.4	25%	0.12	0.031	84%	26%	0.13	33,264	0.11	24.4	4.6%	7.4	+0.94 +- 0.09
E7 crew pool +20%	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
E7 loading sidings +1	5.16 +- 0.03	37.2 +- 0.2	43.4	22%	0.18	0.028	50%	23%	0.13	33,299	0.26	23.5	5.1%	6.3	+0.39 +- 0.05

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
no through goods on main line (A-24 = 0)	4.78 +- 0.05	40.1 +- 0.4	47.1	21%	0.09	0.012	81%	22%	0.06	33,462	0.02	23.2	5.2%	5.2	+0.01 +- 0.07

Proposal scenarios (submitted deck): dispatch, crew booking, exam trigger

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
baseline (loaded first, crew when ready, exam 1 day left)	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
Dispatch: alternating directions	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
Dispatch: batches of 3	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
Crew ordered during loading (A-29)	5.77 +- 0.03	33.3 +- 0.2	38.3	25%	0.04	0.031	89%	26%	0.13	33,440	0.00	24.2	4.6%	7.9	+1.00 +- 0.06
Exam trigger: at expiry	4.80 +- 0.04	40.0 +- 0.3	46.9	21%	0.09	0.026	80%	22%	0.13	33,446	0.02	23.2	5.5%	5.3	+0.03 +- 0.07
Exam trigger: 3 days left	4.75 +- 0.04	40.4 +- 0.4	47.4	21%	0.09	0.025	79%	22%	0.13	33,320	0.05	25.2	5.7%	5.3	-0.02 +- 0.06
Exam trigger: 5 days left	4.78 +- 0.04	40.1 +- 0.4	46.9	21%	0.09	0.026	79%	22%	0.13	33,304	0.02	27.3	6.0%	5.3	+0.01 +- 0.07
Exam trigger: 7 days left	4.75 +- 0.03	40.3 +- 0.3	47.7	21%	0.09	0.027	80%	21%	0.13	33,387	0.05	29.5	6.5%	5.2	-0.02 +- 0.06

Line doubling vs best operating policy (free levers)

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
baseline (single line)	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
Doubling only (Rs 228 cr)	4.88 +- 0.05	39.4 +- 0.4	45.9	11%	0.07	0.026	79%	22%	0.13	33,259	0.01	23.3	5.4%	5.4	+0.11 +- 0.06
Best free policy: crew ordered during loading	5.77 +- 0.03	33.3 +- 0.2	38.3	25%	0.04	0.031	89%	26%	0.13	33,440	0.00	24.2	4.6%	7.9	+1.00 +- 0.06
Best free policy + doubling	5.78 +- 0.03	33.2 +- 0.2	38.3	13%	0.02	0.031	91%	26%	0.13	33,354	0.00	23.7	4.5%	7.8	+1.01 +- 0.07

Assumption sensitivity

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
baseline	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
A-01 GGS loops 3	5.77 +- 0.06	33.3 +- 0.3	39.1	25%	0.03	0.032	81%	26%	0.13	33,222	0.00	23.4	4.5%	7.8	+1.00 +- 0.09
A-02 TAE loops 3	4.54 +- 0.04	42.2 +- 0.4	48.4	20%	0.47	0.025	95%	20%	0.13	33,281	2.43	23.3	5.8%	4.4	-0.23 +- 0.08
A-13/A-14 yard handling halved	5.50 +- 0.06	34.9 +- 0.4	41.2	24%	0.06	0.031	88%	25%	0.13	33,400	0.00	23.7	4.8%	7.0	+0.73 +- 0.08
A-15/A-16 CSTPS handling halved	4.81 +- 0.04	39.8 +- 0.3	46.9	21%	0.09	0.027	80%	22%	0.13	33,432	0.01	23.1	5.4%	5.2	+0.04 +- 0.07
A-24 through goods 40/day	4.80 +- 0.03	39.9 +- 0.2	46.5	21%	0.10	0.045	77%	22%	0.22	33,637	0.05	23.6	5.1%	5.3	+0.03 +- 0.06
A-23 absolute block TAE-CD + A-24 40/day	4.77 +- 0.03	40.2 +- 0.3	46.5	21%	0.10	0.067	75%	21%	0.34	33,379	0.06	23.4	5.1%	5.2	+0.00 +- 0.06
A-21 lateness mean 60 sd 60 min	4.77 +- 0.05	40.2 +- 0.4	47.1	21%	0.09	0.026	79%	22%	0.13	33,367	0.05	24.4	5.5%	5.2	+0.00 +- 0.09
A-25 crew pool 10	4.77 +- 0.06	40.3 +- 0.5	47.1	21%	0.09	0.026	80%	21%	0.13	33,328	0.02	24.3	5.5%	5.0	+0.00 +- 0.00
A-27 controller does not hold empties	4.56 +- 0.06	42.0 +- 0.5	48.9	20%	0.50	0.025	95%	21%	0.13	33,374	1.12	23.7	5.7%	4.7	-0.21 +- 0.07

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
CR-02 no duty extension	4.76 +- 0.06	40.3 +- 0.4	47.4	21%	0.09	0.025	80%	22%	0.13	33,451	0.26	24.8	5.5%	5.0	-0.01 +- 0.04

Combinations to reach LD-01 (7-8 rakes/day)

case	rakes/day	cycle h	P90 cycle h	line util	line queue	Tadali queue	siding occ	tippler util	DN merge wait h	demurrage Rs/rake	crew expiries /100 trips	exam h/rake/month	BPC share	stock min days	delta rakes/day vs baseline
crew at loading + loops +1	6.33 +- 0.04	30.2 +- 0.2	35.7	28%	0.07	0.035	89%	28%	0.13	33,482	0.00	23.7	4.1%	9.0	+1.57 +- 0.06
crew at loading + loops +1 + RLS	8.13 +- 0.03	23.6 +- 0.1	27.3	35%	0.11	0.043	33%	37%	0.13	4	0.00	24.0	3.2%	10.4	+3.36 +- 0.06
loops +1 + Chandrapur lobby	5.99 +- 0.06	32.0 +- 0.4	38.5	26%	0.09	0.034	87%	27%	0.14	33,315	0.00	23.9	4.4%	8.4	+1.22 +- 0.09
loops +1 + lobby + RLS	7.63 +- 0.03	25.1 +- 0.1	28.8	33%	0.15	0.043	33%	35%	0.14	1	0.00	24.2	3.4%	10.3	+2.86 +- 0.06
loops +1 + lobby + RLS, fleet 10	9.13 +- 0.03	26.2 +- 0.1	30.2	40%	0.33	0.052	44%	41%	0.14	1	0.00	24.5	3.6%	10.5	+4.36 +- 0.06
loops 3/3, sidings 3, lobby, fleet 10	7.81 +- 0.03	30.7 +- 0.1	35.4	34%	0.20	0.044	74%	35%	0.13	33,357	0.00	24.5	4.2%	10.1	+3.04 +- 0.06
loops 4/3, sidings 3, lobby, RLS, fleet 12	11.17 +- 0.05	25.8 +- 0.1	29.4	49%	0.28	0.062	32%	50%	0.13	30	0.00	25.7	3.5%	10.8	+6.40 +- 0.06

Baseline cycle-time decomposition (phase 3, mean h per cycle)

stage	mean h	share
Placement at GGS (A-13)	1.82	5%
Wait for loading siding	1.78	4%
Loading (LD-03/04)	6.32	16%
Release->ready GGS (A-14)	2.77	7%
Wait for crew (loaded)	2.76	7%

stage	mean h	share
Wait single line + TAE loop	0.04	0%
Run GGS->TAE spur	0.52	1%
Wait DN path at TAE	0.13	0%
Run TAE->CD (loaded)	0.33	1%
Placement at CSTPS (A-15)	3.50	9%
Wait for tippler	0.00	0%
Unloading (UL-02)	4.31	11%
Release->despatch CD (A-16)	4.15	10%
Wait for crew (empty)	1.68	4%
Held at CD: loops/UP path	7.54	19%
Run CD->TAE (empty)	0.33	1%
Wait single line at TAE	0.40	1%
Run TAE->GGS spur	0.52	1%
Crew relief (duty expired)	0.00	0%
BPC exam detour	1.37	3%
total cycle	**40.3**	